DATA ITEM DESCRIPTION

Title: Semiconductor Process Specification

Number: DI-QCIC-80924A
Approval Date: 08 NOV 2006
AMSC Number: 7631
Limitation: N/A
DTIC Applicable: N/A
GIDEP Applicable: None
Office of Primary Responsibility: NS/DA023
Applicable Forms: N/A

Use/relationship:

The Semiconductor Process Specification describes the electrical parameters/characteristics and the topological design rules (artwork layout) required for the fabrication of semiconductor integrated circuits. It includes information on the derivation of these parameters, and the testing requirements to verify accuracy. The data is used as a procurement specification to assure that equivalent circuit performance can be obtained for suppliers.

This Data Item Description (DID) contains the format and content preparation instructions for each tech data product generated by the specific and discrete task requirement as delineated in the contract.

This DID is used on systems/equipment acquisitions where semiconductor integrated circuits are designed and fabricated.

This DID supersede DI-QCIC-80924

Requirements:

1. Reference documents: None.

2. Format: The data shall be prepared in contractor format on 8 ½ x 11" (metric size A4) bond paper. Drawings and illustration fold-outs may be used, but must not exceed 8 ½ x 11" limits when folded. Photo-reduction of oversized pages is preferred, provided such reductions are easily readable and reproducible.

3. Content. The report shall contain the following information:

3.1 Cover/title page. The following information shall appear on the cover and title page:

a. Title
b. Sequence number
c. Contract number
d. Contractor name and address
e. Processing facility name and address

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f. Report name.

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4. Body of Report.

4.1 Electrical Parameters/characteristics. For each of the design parameters listed in Table 1, the following shall be specified:

4.1.1 Parameter value/characteristics in the form of histograms, distributions, minimum/maximum or typical values.

4.1.2 The sample size upon which the values were based, where applicable.

4.1.3 Information on parametric circuits used to determine and quantify the parameters.

4.1.4 The test set-up procedures, and conditions under which the parameters values were determined.

TABLE 1. Electrical Parameters

Parameter	Parameter
1. Starting material	9. Junction breakdowns
2. Diffusion resistivity	10.Doping concentrations
3. Mobility	11. Body effects
4. Oxided thickness	12. Punch-through voltages
5. Diffusion depth	13. Metal resistances
6. Gate and field thresholds	14. Delta threshold limits under stress biasing
7. Gain factors	15. Other as applicable
8. Capacitances	

4.2 Topological design rules. This section describes the Topological (artwork layout) Design rules to be followed in the layout of the fabrication process. For each applicable areas listed below, a pictorial representation of the layout rule shall be Included reflecting associated minimum, maximum, or typical values sufficient To permit layout or modification of the circuit layout.

a. Line widths.
b. Line spacing.
c. Transistor device geometric
d. Contact design for level-to-level interconnect
e. Diffusion widths
f. Diffusion-diffusion spacing/overlap
g. Bonding pad sizes and spacing to scribe line and circuit areas.
h. Scribe area description.
i. Alignment marks

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j. Standard test devices

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k. Numerical mask and artwork level identification including:
    1. level nomenclature
    2. Fiduciary markings
    3. Coloring/crosshatching used to distinguish levels
    4. Polarity of each mask level

5. END OF DI-QCIC-80924A

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